Interactive Explanations for Reinforcement Learning: Taking Advantage of Progressive Disclosure

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Abstract. In recent years, Artificial Intelligence (AI) has come to the forefront of society, due to various advancements in the field of Machine Learning (ML), such as the rising power of Large Language Models (LLMs). Nowadays, people of all skill levels are in contact with AI, from novice users to experts. One of the three ML paradigms, Reinforcement Learnings (RLs), has been responsible for much of this progress, playing a key role in the training of LLMs and surpassing humans in various tasks. RL involves an agent interacting with an environment and learning through trial-and-error. As the agent selects which action to take for each situation (state), it is rewarded accordingly to its choice, allowing it to improve its performance at the task at hand. However, the great capabilities of modern AI have brought along with them tremendous complexity; as a result, even AI experts cannot fully comprehend these approaches. Due to AI's pivotal role in the future of mankind, the topic of trustworthy AI has gained a lot of attention recently. Trustworthy AI covers the key principles that need to be respected when working with this technology. One of such principles is transparency, which involves making AI more understandable to humans and is the main focus of Explainable Artificial Intelligence (XAI). In spite of the research community making strides in this area, various challenges remain. For instance, transparency by itself seems insufficient to attain human trust and for many years researchers tended to ignore non-expert users in their studies. Furthermore, research intersecting XAI and social sciences has argued that AI explanations should work similarly to a conversation, allowing for a dynamic back-and-forth between the system and the user. After all, such an interaction leads to a gradual and natural understanding of the inner workings of the AI. Despite positive reception of this theory, the vast majority of AI explanations remain static to this day. On top of that, works related to interactive explanations tend to neglect RL, with only a couple works on this intersection existing. Therefore, we propose an interactive XAI approach for RL that leverages concept-based explanations. Our solution takes advantage of progressive disclosure, a paradigm that consists of revealing information to the user on demand to meet their needs and avoid overwhelming them with unnecessary details. While prior work has used this principle to some extent, we make it a cornerstone of our contribu-

tion. Additionally, we take a completely different approach that follows a theoretical framework developed by Weld and Bansal. For the sake of comparison, we built two versions of our XAI system, a static one and an interactive one. We hope that our contribution paves the way towards more research that combines interactive explanations with RL.

Keywords: Explainable AI · Reinforcement Learning · Interactive Explanations.

1 Introduction

The importance of AI [15] keeps growing in our modern world. Thanks to rapid advancements in the field, its use has become intertwined with society. Nowadays, not only is it being used by AI experts, but also by the average person, who is now much closer to these technologies. These progresses have been in large part due to the increasing power of LLMs, which employ RL at certain points during their training. In RL [21], we consider an agent that learns by interacting with an environment. The agent gradually picks actions that change the environment's state and gets rewarded according to how beneficial (or not) the change in the state was. This way, it is possible to incrementally improve the agent's policy, which is how it decides which action to select in each state.

Unfortunately, the growing complexity of AI algorithms is a double-edged sword. As much as it may have catapulted AI to the forefront of society, it has also led to problems of understandability. More concretely, the Neural Networks (NNs) used are incomprehensible by even AI experts, let alone the average person. How can we deploy these algorithms in real life if we don't comprehend how they work? And it is not just an issue of transparency either; notably, other issues such as fairness, privacy, and accountability are also of great concern. These concerns have led to the rise of trustworthy AI [8,9,12,22], a research field that concerns itself with these kinds of topics. Within this area, there is a subtopic dedicated to explainability, XAI [3], which has gained a lot of traction in recent years.

Despite these advancements, many issues still remain. For instance, some studies have shown that trust did not improve even with better understandability [11,13]. Therefore, measuring trust is something that should be considered of utmost importance [10]. Additionally, until more recently, the average user who lacks technical knowledge tended to be neglected. Since they are often the ones who receive the explanations, it is clear that more care needs to be taken with regard to them. Lastly, various papers have argued that explanations should be interactive, since explanations between humans are ultimately a social interaction (conversation). As such, users should be more receptive to explanations that allow them to gradually probe the information, which is akin to asking follow-up questions.

Thankfully, these kinds of explanations are becoming more common. For example, Weld and Bansal [25] propose a theoretical framework for handling such explanations (see Figure 7 from their work). At the same time, progressive

disclosure [19,5], or gradually showing users the information as they interact with the system, is being suggested as a way to better adapt to users' needs. From a more practical standpoint, the number of works in the area that are tested with human subjects has increased as well [18,17], with a survey about this topic even gaining traction in the research community [4]. However, these works are still the exception and not the norm. Furthermore, this kind of approach has mostly been applied in the context of supervised learning, with only a few works on RL gathering interest from the academic community [1,6,14]. Considering how RL has seen multiple successes in recent years, it seems only natural to try to apply this explanatory methodology to it. Moreover, we stress the use of progressive disclosure in our work. While these works do use this design principle to some extent (some more [14] than others [1,6]), we make sure that the entire XAI system abides by this principle. Additionally, none of the three works surveyed adopt an approach similar to what is envisioned by Weld and Bansal [25], which we do.

1.1 Research Aim

With our research, we aim to develop and test interactive explanations for RL, in the molds of what is suggested by Weld and Bansal and by taking advantage of progressive disclosure.

We now present the research objectives.

1. Develop explanations for RL that allow the user to query progressively for more information.
2. Test this explanatory approach using LLM-as-a-Judge, evaluating explanation goodness, satisfaction, and trust.
3. Test our XAI system with human subjects, evaluating those same criteria.

As for the research questions, they are the following:

1. What kind of Explainable Reinforcement Learning (XRL) method is suited to be the basis of our approach?
2. How do the artificial users from the LLM-as-a-Judge methodology evaluate our XAI system?
3. How do average users with no technical knowledge of AI feel about the system developed?

1.2 Document Overview

In Section 2 we cover literature related to interactive explanations for RL. As for Section 3, it contains the methodology used, including the RL environment developed and the heuristics used for comparison, the concept-based XRL approach employed, the static version of our explanatory system, and the interactive version we propose. In Section 4, we go over the experimental setup, results, and discussion of the three experiments we conducted (comparison of performance against simple heuristics, evaluation using LLM-as-a-judge, and a study with human users). Lastly, we present the conclusion in Section 5.

2 Related Work

Cruz and Igarashi [6] worked on an interactive XRL approach in which the user can interact with a video of the agent playing the game. From there, it is possible to select a portion of the video and pause it, with the agent communicating to the user various kinds of information about that moment. Then, the person receiving the explanation (the explainee) can ask why (not) an action was selected. Not only that, but using dropdown menus, it is possible to change the agent's behavior. Meanwhile, Amitai et al. [1] created an approach that allows users to see videos of the agent playing, after defining what kind of situations they want to see. This approach allows the explainee to iteratively get a better grasp of why the agent is behaving a certain way. While both of these works feature some progressive disclosure, it is not for the whole XAI system. Also, their approaches are focused on interaction with videos, which is quite different from our methodology (see Section 3).

Mishra et al. [14] developed an approach that allows the user to visualize various kinds of information about the whole state space, such as actions, rewards, policies, values, among others. The explainee can also ask three types of questions to the agent about its action choice: why, why not, and when. The authors made sure to tailor their methodology to non-experts in RL. Their approach clearly employs progressive disclosure, even if it is not explicitly mentioned. However, this visualization of the whole state space is different from our methodology, in which we allow the user to inspect each "state" at a time and make the final decision (in reality, we use a "state" composed of multiple states, as we will explain later).

3 Methodology

3.1 The Environment and the Heuristics

The Environment. For the RL environment, we selected the "haulier capacity matching" problem. Transportation problems [2,20] can be subdivided in various specific issues, one of which is the pickup and delivery problem [26]. This problem involves (1) accepting (rejecting) an order, (2) assigning an order to a vehicle, and (3) picking the best route for the vehicle. When we say "haulier capacity matching" we are referring to these first two steps. We stress that in [24], it is underscored that RL could be employed for more complex scenarios, further justifying our decision of selecting this domain.

More specifically, the variables we are concerned with and form our state are: the distance travelled, the distance travelled with no freight, and the grams of CO2 emitted. This last one is particularly important because as mentioned in [26], only 12% of the papers surveyed took sustainability and environmental concerns into account.

From there, the scenario considered is identical to the one used in [16]. We will now describe said scenario.

Our environment consists of 125 trucks and 250 freight (orders). These trucks and freight are generated randomly each time the algorithm runs. Additionally, they are heterogeneous, which means they each truck needs to be matched with the appropriate freight. We used [3] as a reference for the categories ("DryCargoTruck", "RefrigeratedCargoTruck", and "SpecialCargoTruck") and types of trucks, which depend on the containers they can hold. In particular, these categories and types dictate the truck's maximum payload (in Kg) and length, width, and height (in mm).

As for the trucks, they also have between 3 and 7 availabilities, which are the periods of time in which they can perform pickup and delivery. Additionally, the trucks have randomly generated emission rates (between 58.3 and 307.2, in grams of CO2 per ton-kilometre, gCO2/t-km), to match with the values from [4]. Finally, every truck is assigned a current location.

At the same time, the freight are characterised by their category ("Dry", "Refrigerated", and "Special"), weight, length, width, and height. Furthermore, each cargo has an arrival and deadline, and an origin and destination.

The time frame considered is one randomly generated week; that is to say, the 250 freight are to be assigned to the availabilities of the 125 trucks during that week. That being said, we go through each order individually, beginning with the ones from the first day of the week and moving along day by day.

The locations are based on a simple distance grid with predefined locations. The data was created using Google Maps as a reference in terms of distances and travel time. Nine cities from a European country were chosen for this purpose.

For the RL approach, we took advantage of Python's gymnasium library [23].

The Heuristics. Once again, the heuristics are the same as the ones used in [16].

Heuristic_1 simply matches the freight with the first available truck it finds. This is essentially the most basic approach to solve the problem and is used as a baseline for what the minimum performance should be.

As for heuristic_2, for each freight it runs through all the trucks and checks which ones are available to deliver it. Then, it picks the "best" one according to its own criteria (which depends first on the distance travelled with no freight and second on the grams of CO2 emitted). This heuristics is still very simple, but it attains much better results than heuristic_1.

3.2 The Explainable Reinforcement Learning Approach used as the Basis

To make the approach explainable, we experimented with different explainability approaches to see which ones could work. We first looked to Local Interpretable Model-Agnostic Explanations (LIME) and SHapley Additive exPla-

[3] https://www.maersk.com/industry-sectors/dry-cargo

[4] https://theicct.org/publication/co2-emissions-from-trucks-in-the-eu-an-analysis-of-the-heavy-duty-co2-standards-baseline-data/

nations (SHAP), which are two of the most popular methods. Unfortunately, neither of them seemed to work well for our scenario.

For LIME, the issue was that most of the explanations did not make sense. For example, in Figure 1 it predicts a bad match with 100% probability, but the explanation itself makes it seem like a good match instead; moreover, it is classifying having higher values as a good thing, which makes no sense in this context.

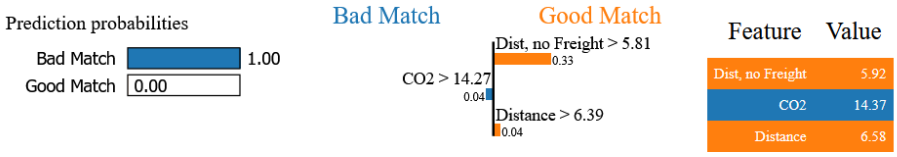


Fig. 1. LIME giving a bad explanation

As for SHAP, we tried to use Kernel SHAP, but quickly realized it would not be adequate due to its assumptions; notably, Kernel SHAP assumes independence, which is not the case here. For example, the distance travelled depends on the distance travelled with no freight: it will always be the distance travelled with no freight plus the distance travelled with freight.

Due to the most popular approaches being unsuited for the problem, we searched the literature for a different method. Ultimately, we decided to use State2explanation [7], in part due to having its code available to the public. This method attaches concepts to states, making the results interpretable at a higher level of abstraction. As a result, it is easier for humans to understand why a certain choice was made by the AI.

For our work, we designed the concepts around thresholds for the three main metrics: distance travelled, distance travelled with no freight, and grams of CO2 emitted. The thresholds are 427 Km, 166 Km, and 1,200,000 grams, respectively. We decided on these thresholds after examining multiple outputs, checking which values were possible and what could be considered low or high. These thresholds are then used to map the concepts, which themselves correspond to explanations, as seen in Table 1.

During the process of creating the dataset, we collect 20,000 impossible matchings and 130,000 possible matchings, for a total of 150,000. From there, we create a misaligned dataset, with $z = 8$ misaligned concepts (since we have nine concepts in total, one will be aligned, and eight will be misaligned). For more information about how the aligned and misaligned concepts work, check [7]. In total, the misaligned dataset has 1,200,000 matchings, which are the initial 150,000*8 misaligned concepts. Lastly, we join the datasets and shuffle the data.

As for how the concepts are used during the RL process, essentially the reward depends on the predicted concept and the action selected for each given

Table 1. Mapping of concepts to explanations.

Concept	Explanation
null	impossible matching
SD_SDF	short distance travelled and short distance travelled with no freight
SD_LCO2	short distance travelled and low CO2 emissions
SDF_LCO2	short distance travelled with no freight and low CO2 emissions
SD_SDF_LCO2	short distance travelled, short distance travelled with no freight, and low CO2 emissions
LD_LDF	long distance travelled and long distance travelled with no freight
LD_HCO2	long distance travelled and high CO2 emissions
LDF_HCO2	long distance travelled with no freight and high CO2 emissions
LD_LDF_HCO2	long distance travelled, long distance travelled with no freight, and high CO2 emissions

state. Each state is defined by the distance travelled, distance travelled with no freight, and grams of CO2 emitted. Meanwhile, action 0 corresponds to the agent considering the current matching a bad choice, while action 1 is the agent defining this matching as a good choice. We can see how the rewards are attributed in Table 2.

Table 2. Mapping of actions and concepts to rewards.

Action	Concept	Reward
0	null	0.01
0	SD_SDF or SD_LCO2 or SDF_LCO2	0
0	SD_SDF_LCO2	0
0	LD_LDF or LD_HCO2 or LDF_HCO2	20
0	LD_LDF_HCO2	30
1	null	0
1	SD_SDF or SD_LCO2 or SDF_LCO2	20
1	SD_SDF_LCO2	30
1	LD_LDF or LD_HCO2 or LDF_HCO2	0
1	LD_LDF_HCO2	0

3.3 Static Explanations

We used a static version of our system as a baseline to compare to during experiments. We will now go through its screens. We note that we used the Python library streamlit for the user interface.

In figure 2, we see the first screen. In it, we can see the truck-freight matchings accepted by the AI system, with their distance travelled, distance travelled with

no freight, and grams of CO2 emitted being shown. By clicking the "Switch" button, the user can instead see the truck-freight matchings rejected by the AI, which corresponds to the screen in figure 3.

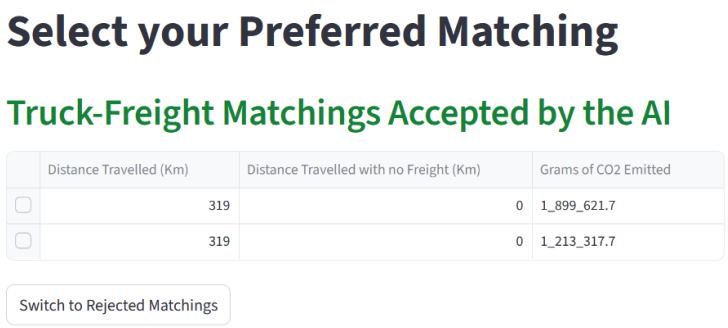


Fig. 2. Truck-Freight Matchings Accepted by the AI

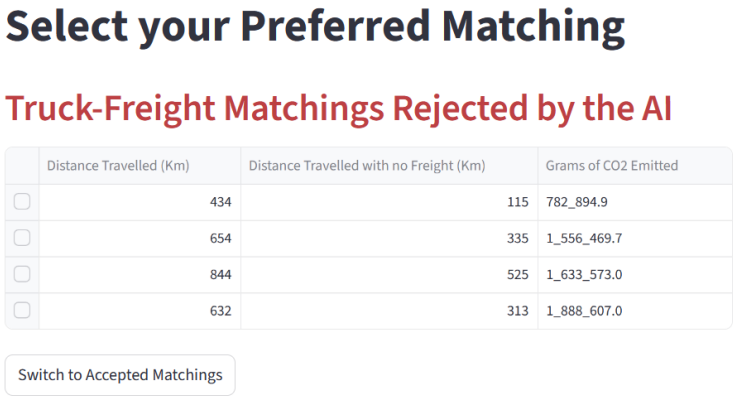


Fig. 3. Truck-Freight Matchings Rejected by the AI

Once the user clicks on a specific matching, they are taken to the screen shown in figures 4 and 5, which is a single screen the user can scroll through.

In figure 4, besides the basic information about the matching, it is also possible to see information about the travel time, the truck company, and also the locations of the truck, the pickup, and the destination. Afterwards, the status of the matching is shown as either accepted or rejected, with an explanation being given as to why that is so. This explanation is based on the concepts the model trained on, as explained in subsection 3.2.

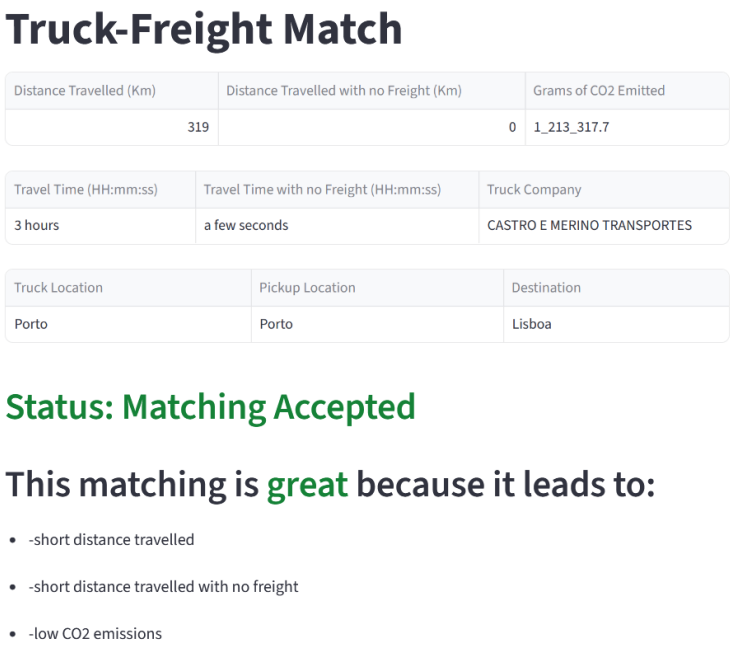


Fig. 4. Information about a matching (Part 1)

In the second half of the screen, figure 5, we first get a generic explanation as to how the model gets to its decision, which involves telling the user about the number of scenarios the model was trained on and how many of them are similar to the one selected. From there, three matches the model was trained on that are similar (same concept) are shown to the user. Lastly, the user can either select this matching or go back to the initial screens from figures 2 and 3 to ponder over their decision.

Obviously, this is a lot of information to present to the user in one single page, which then leads us to the interactive version of the approach, which is sectioned into multiple screens.

3.4 Interactive Explanations

We will now present the screens the user is taken through as they interact with the interactive version of the system. As you look through the figures, notice how instead of overwhelming the user with information, the varied aspects of explanations are shown in different screens. This configuration lines up with the principle of progressive disclosure. Additionally, pay attention to how the flow from screen to screen is based on what is suggested by Weld and Bansal [25].

Since the first two screens are identical to the static version (figures 2 and 3), they will be omitted. We will skip ahead for the moment when the user selects a matching.

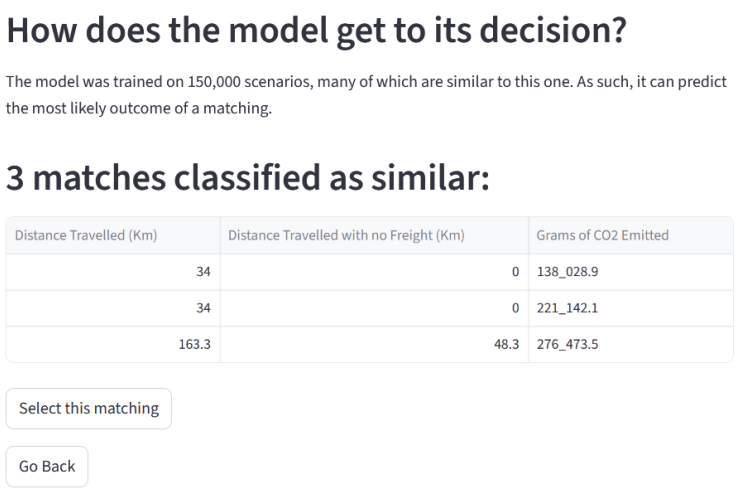


Fig. 5. Information about a matching (Part 2)

In figure 6, we see some additional information about the match selected. Just as in the static version, we can see information about the time taken, the truck company, and the various locations. Moreover, whether the matching was accepted or rejected is highlighted. Additionally, it is possible to select the matching. Where it differs from the static version is in the addition of the "Why did the AI accept?" button, which allows the user to find that out on demand, instead of being overwhelmed with too much information at once (the same happens if the matching was rejected).

After clicking that button, the user is taken to figure 7. In this screen, an explanation for the acceptance/rejection is given based on the concept attached to the corresponding state. On top of that, there is a button "How does the model get to its decision" which can be clicked for additional information.

This leads us to figure 8, in which a generic explanation is given to the user, highlighting the number of scenarios the model was trained on and that many were identical to the one chosen. From there, the user can click a button to find out training examples similar to this one. That way, the user can verify that indeed there were training examples identical to this matching.

We now arrive at figure 9. In this screen, three training examples classified as similar are shown to the user. These training examples correspond to the same concept as the one from the match being considered. At that point, the user might wonder what would happen if a value was different, which can be done through the button "What if a value was different?".

In figure 10, we see inputs for the user to introduce their values, allowing them to verify what the model would do with their custom selection of values. These values are analysed by the model and attached to a concept, giving rise to an explanation in the following screen.

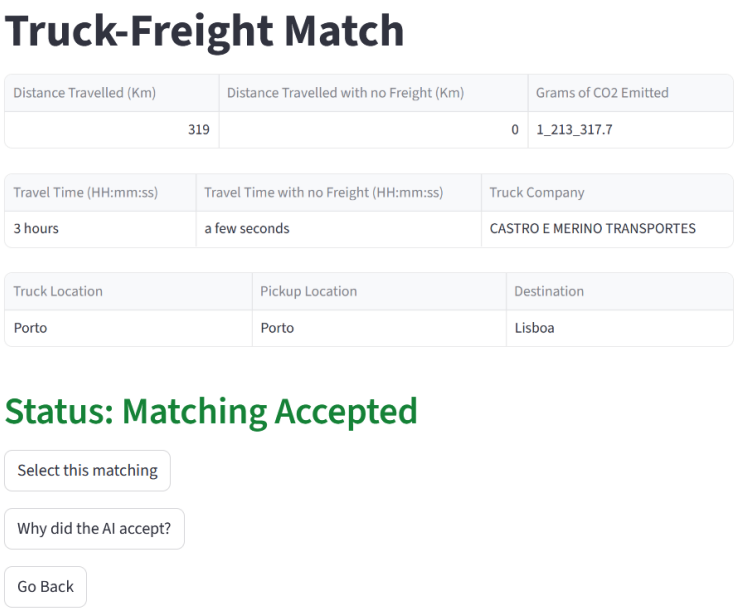


Fig. 6. Information about a Truck-Freight Match

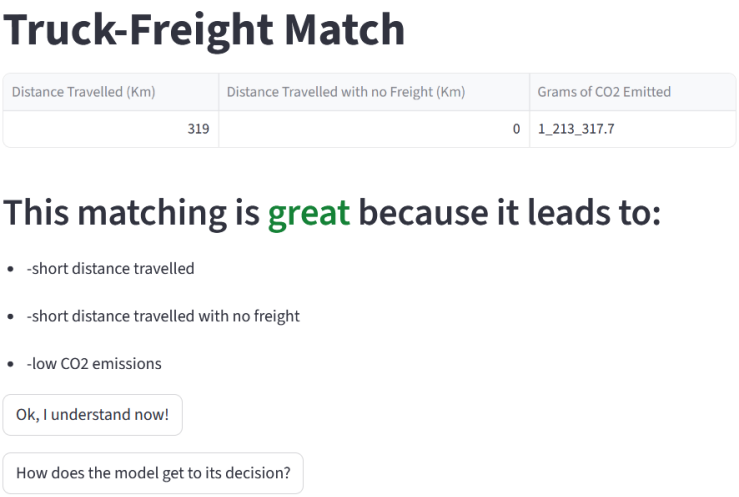


Fig. 7. Why the AI Accepts/Rejects a Match

Lastly, in figure 11, we see an explanation for the acceptance/rejection of the custom scenario created by the user. From here, it is possible to once again go back to the previous page with the button "What if a value was different?", in case the user is not yet satisfied.

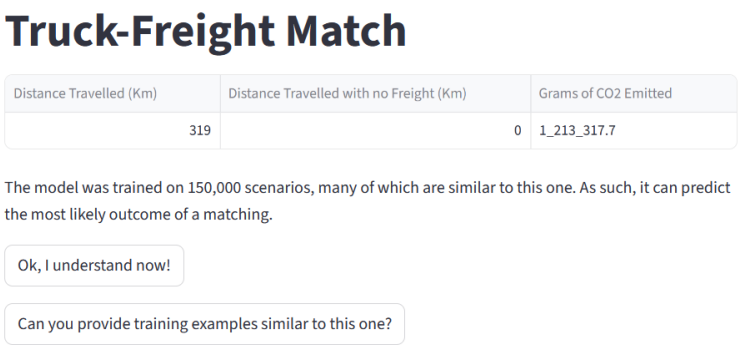


Fig. 8. Generic Explanation of the Decision

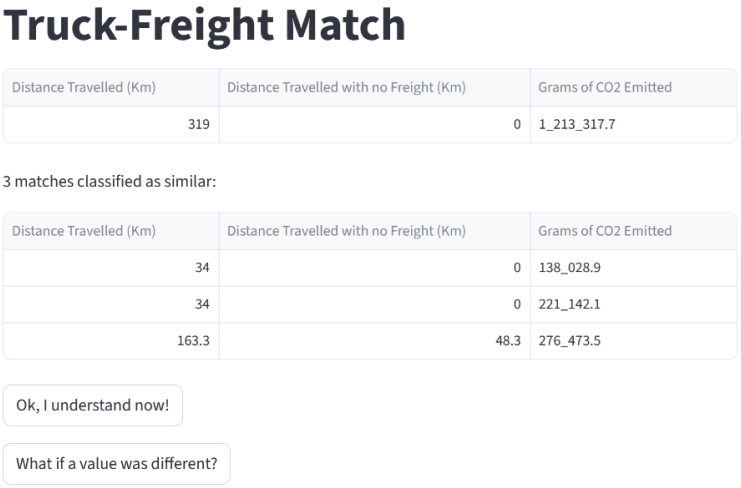


Fig. 9. Training Examples that are Similar

We stress that if at any point the user clicks the "Ok, I understand now!" button, they are taken back to the information screen from figure 6. This way, they avoid having to click buttons multiple times to go back to that screen.

4 Experiments

4.1 Compare Performance with the Heuristics

Experimental Setup. Our experimental setup for comparing with the heuristics is similar to the one from [16]. The main difference is that instead of comparing imitation learning approaches with the heuristics, we are instead comparing a concept-based XRL approach with those same heuristics.

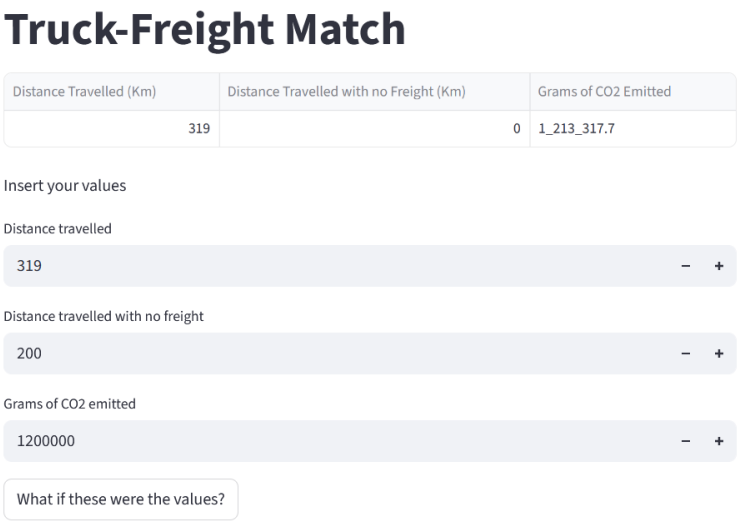


Fig. 10. What if Some Values Were Different?

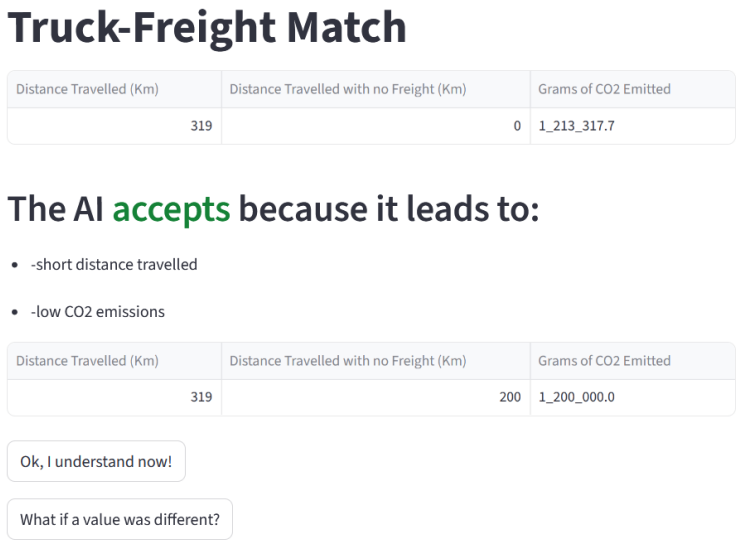


Fig. 11. What if Some Values Were Different? (Part 2)

For each of the approaches, we ran the same setup 30 times, for the sake of statistical significance. The experiment involved keeping track of the percentage of freight delivered, the distance travelled (Km), the distance travelled with no freight (Km), and the grams of CO2 emitted. For the last three metrics, the final results are modified so that it is assumed that every freight was delivered.

Otherwise, algorithms that delivered more freight would be punished for doing so (because their metrics would then be higher).

While for the percentage of freight delivered we simply take the mean and standard deviation, for the other three metrics we run various experiments in R. Just as in [16], the percentage of freight delivered always hovers around 80% regardless of the algorithm, so we focus our efforts on the other three metrics.

Taking into account an $\alpha = 0.05$, we use a one-way ANOVA for the three key metrics: the distance travelled, the distance travelled with no freight, and the grams of CO2 emitted. The null hypothesis is that the metric is identical for all the algorithms. Meanwhile, the alternative hypothesis is that there is at least one pair with significant differences. For each scenario, we generate boxplots and test the assumptions of the ANOVA.

We perform a pairwise t-test, with Holm-Bonferroni correction, whenever the null hypothesis is rejected; that way, we figure out which pairs have statistically significant differences.

Results. In Table 3, we can see the results for the percentage of freight delivered, which was identical for all algorithms; as such, we can deduce that the algorithm does not seem to create a significant difference for this evaluation criteria.

Table 3. Percentage of Freight Delivered.

Heuristic_1	Heuristic_2	RL
80.4 ± 3.3	81.0 ± 2.7	80.1 ± 2.8

The boxplots for the distances travelled are shown in Figure 12. It is clear that the RL approach seems to present the best results, having the shortest distances travelled.

As for the ANOVA test, the assumptions of normality and homoskedasticity were met. The null hypothesis was rejected with a p-value smaller than $2e-16$. As a result, we performed a pairwise t-test. We can see the results in Table 4.

Table 4. Pairwise t-test for the Distances Travelled.

	Heuristic_1	Heuristic_2
Heuristic_2	< **2e-16**	
RL	< **2e-16**	**2.9e-08**

Every null hypothesis was rejected, which means all of the algorithms are considered statistically different from each other. This result matches with what we saw on the boxplot.

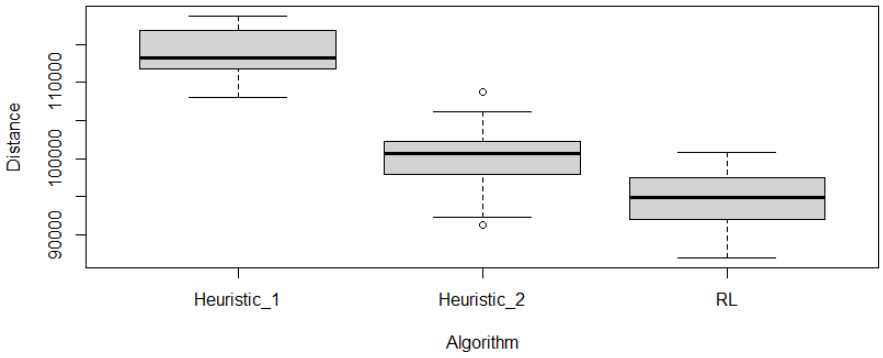


Fig. 12. Boxplots for the Distances Travelled

In Figure 13, we present the boxplots for the distances travelled with no freight. The results are identical to the first boxplot: once again, RL is the best, followed by heuristic 2, with heuristic 1 having by far the worst results.

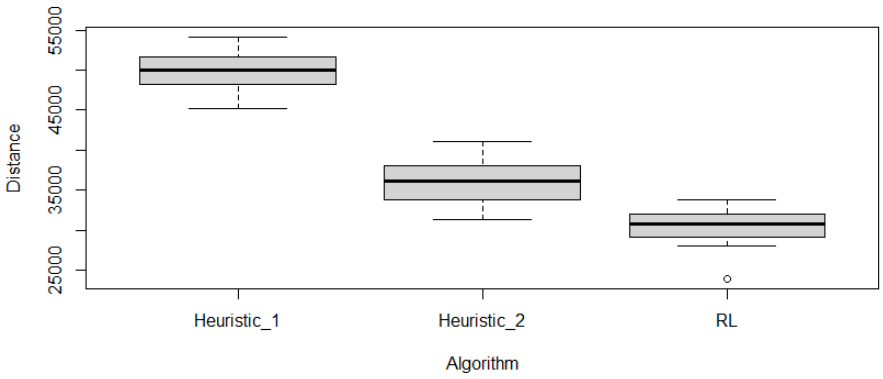


Fig. 13. Boxplots for the Distances Travelled with no Freight

The assumptions were once again met for the ANOVA test. The null hypothesis was rejected with a p-value smaller than $2e-16$. As such, we conducted a pairwise t-test. The results are shown in Table 5.

Table 5. Pairwise t-test for the Distances Travelled with no Freight.

	Heuristic_1	Heuristic_2
Heuristic_2	< **2e-16**	
RL	< **2e-16**	**5.4e-13**

These results are also similar to the ones for the distances travelled. Once more, every null hypothesis is rejected, with every pair of algorithms being considered statistically different.

In Figure 14 we can visualise the boxplots for the grams of CO2 emitted. This time, heuristic 2 seems to have the best results, with the other two being somewhat similar.

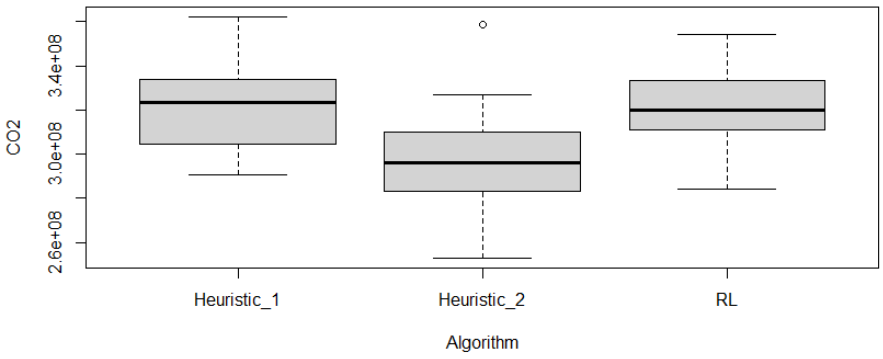


Fig. 14. Boxplots for the Grams of CO2 Emitted

Once again, the assumptions of normality and variance of the residuals are met. The null hypothesis was rejected with a p-value of $2.43e-06$. Therefore, we performed a pairwise t-test. Table 6 shows the results of that test.

Table 6. Pairwise t-test for the Grams of CO2 Emitted.

	Heuristic_1	Heuristic_2
Heuristic_2	**2.1e-05**	
RL	0.95	**2.1e-05**

The results from Table 6 show us that heuristic 1 and RL are considered to be statistically similar, which is in line with the results from the boxplots.

Discussion. While our approach outperforms the heuristics for both the distance travelled and distance travelled with no freight metrics, its results are more underwhelming for the grams of CO2 emitted. More specifically, it seems to have about the same performance as heuristic 1, which is an incredibly simple heuristic. These results highlight the complexity of the problem, with it being difficult to balance the different metrics. Therefore, we can conclude that we should strive to make our approach better in terms of balancing the CO2 emitted with the distance travelled metrics.

4.2 Tests using LLM-as-a-Judge

Note: this subsection is not finished.

Experimental Setup.

Results.

Discussion.

4.3 Tests with Human Subjects

Note: this subsection is not finished.

Experimental Setup.

Results.

Discussion.

5 Conclusion

Note: this section is not finished.

Disclosure of Interests. The authors declare that they have no competing interests relevant to this work.

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